

DevDash'26 – Hackathon Project Report

OneUni powered by UCL

A Centralised Student Information & Campus Services Platform

Team 404 : Team Not Found

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Competition: DevDash'26

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GitHub Repository: <https://github.com/DevDash-26/404-Team-Not-Found>

1. Executive Summary

UCL one uni is a centralised digital platform designed to provide students with one reliable location for accessing university-related information and services.

Currently, students may need to rely on different channels such as WhatsApp groups, notice boards, lecturers, societies and word of mouth to obtain information. This can make it difficult to find accurate and up-to-date information.

Our proposed solution brings important student services into one platform. The prototype provides access to announcements, events, societies, classroom bookings, lost and found services, job and internship opportunities, frequently asked questions and an AI-powered assistant.

The solution was designed with students as the primary users while also providing authorised staff and society representatives with appropriate content-management capabilities.

2. Problem Understanding

The DevDash'26 problem statement identifies a gap in the way students access campus information.

Students currently depend on several disconnected communication channels. These include WhatsApp groups, lecturers, notice boards, societies and word of mouth. As a result, there is no single channel that students can consistently treat as the authoritative source of university information.

This can result in:

- Difficulty finding important announcements
 - Students missing events and opportunities
 - Repeated questions being directed to administrative staff
 - Difficulty finding lost and found items
 - Difficulty accessing campus services
 - Information being distributed through different channels
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3. Project Objectives

The main objectives of One Uni powered by UCL are:

1. Provide students with a centralised source of campus information.
2. Improve visibility of university events and student activities.
3. Reduce the time students spend searching for information.
4. Reduce repetitive enquiries received by administrative staff.
5. Provide convenient access to student services.
6. Provide authorised staff with tools to maintain information.
7. Create a solution that can potentially be extended beyond the hackathon.

These objectives align with the broader outcomes identified in the DevDash'26 problem statement, including a more informed student body, greater visibility of campus activities and reduced time spent searching for basic information.

4. Proposed Solution

We developed **UCL**, a centralised web platform for students and university staff.

The platform consists of two main user areas:

Student Portal

Students can:

- View announcements
- Browse events
- Express interest in events
- Discover societies
- Find lost and found items
- Request classrooms
- Find academic support
- View FAQs
- View academic dates

- Find jobs and internships
- Access library information
- Access IT support information
- Use the AI assistant

Staff/Admin Portal

Authorised users can:

- Publish announcements
- Manage events
- Manage society information
- Update FAQs
- Manage lost and found information
- Manage classroom availability
- Maintain other campus information

UX Design: [OneUni — Master UX Architecture](#)

5. Requirements Coverage

The following table shows the business requirements addressed by our prototype.

Requirement	Implemented Feature	Status
Unified Access	Central student dashboard	Implemented
Targeted Announcements	Faculty/year-based announcements	Implemented
Event Visibility	Events section	Implemented
Event Interest	Event interest button	Implemented
Society Visibility	Society directory	Implemented
Lost & Found	Lost & Found section	Implemented
Classroom Booking	Room availability and booking	Implemented
FAQ Access	FAQ section	Implemented
Access Levels	Student/staff roles	Implemented
Academic Calendar	Academic calendar	Implemented
Emergency Communication	Emergency announcements	Implemented
Job & Internship Visibility	Opportunities section	Implemented
Staff Directory	Staff directory	Implemented
Library Resources	Library information	Implemented
AI Assistant	AI-powered assistant	Implemented

6. Non-Functional Requirements

6.1 Usability

The interface was designed to be simple enough for first-time student users to understand without extensive training.

We used:

- Clear navigation
- Simple terminology
- Consistent buttons
- Categorized information
- Search functionality

6.2 Performance

The application was designed to provide quick responses when navigating between sections.

We reduced unnecessary data requests and organized information into separate sections to improve responsiveness.

6.3 Security & Privacy

Different access levels were implemented for students and authorized staff.

Students can access information intended for them, while administrative functions are restricted to authorized users.

This addresses the security and privacy expectations described under NFR4.

6.4 Maintainability

The application was organized into separate components/modules so that future developers can modify or extend individual features without significantly affecting the rest of the system.

7. Target Users

7.1 Students

Students are the primary users of the system.

Their main requirements include:

- Finding university information
- Viewing announcements
- Finding events
- Finding societies
- Booking classrooms
- Finding opportunities
- Getting answers to common questions

7.2 Staff

Staff users are responsible for maintaining official university information.

They can:

- Publish information
- Update information
- Manage events
- Respond to student needs

7.3 Society Representatives

Society representatives can maintain information about their society and communicate activities to students.

8. Technology Stack

Component	Technology
Frontend	TypeScript
Backend	Java Script
Database	Firebase
Version Control	GitHub
Design	Figma

The technologies were selected based on development speed, maintainability and suitability for the prototype.

9. Key Features & Implementation

9.1 Centralised Dashboard

The dashboard acts as the main entry point to campus information.

Instead of requiring students to search multiple communication channels, the platform provides access to several services from one location.

9.2 Event Management

Students can view upcoming events and their details.

The event page includes:

- Event title
 - Date
 - Time
 - Location
 - Description
 - Organiser
 - Interest registration
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9.3 Lost & Found

Students can report lost or found items.

A submission can include:

- Item name
- Description
- Location
- Date
- Contact information

Users can then search existing entries.

9.4 Classroom Booking

Students can check available classrooms and submit booking requests for study or group work.

9.5 AI Assistant

The AI assistant allows students to ask questions using natural language.

For example:

"Where can I find the library?"

Instead of manually searching through multiple sections, the assistant can guide the student towards the relevant information.

This specifically calls for an AI assistant capable of responding to conversational queries and helping students find information quickly.

10. Security & Access Control

The system uses role-based access control.

Student

Students can:

- View information
- Register for events
- Submit requests
- Report lost/found items

Staff

Authorised staff can:

- Create announcements
- Update information
- Manage events
- Maintain university content

This prevents unauthorised users from modifying official information.

11. Testing

Testing was conducted to ensure that the implemented features operated according to their intended behaviour.

Test ID	Category	Test Case Description	Steps to Execute	Expected Result	Status
TC001	Functional	Unified access to dashboard	Log in as a student and open the home dashboard	All info categories (announcements, events, societies, etc.) are accessible from one screen	Pass

TC002	Functional	Access without login	Attempt to open the app/site without logging in	User is prompted to log in; no campus data shown until authenticated	Pass
TC003	Functional	Targeted announcement delivery	Post an announcement targeted to a specific faculty/year group	Only students matching that faculty/year see the announcement	Pass
TC004	Functional	University-wide announcement	Post a general, university-wide announcement	All students see the announcement regardless of faculty/year	Pass
TC005	Functional	Targeted announcement exclusion	Check a non-matching student's feed after a targeted post	Non-matching students do not see the targeted announcement	Pass
TC006	Functional	University event visibility	Create a university-organised event	Event appears correctly in the events feed	Pass
TC007	Functional	Student-organised event visibility	Create a student/society-organised event	Event appears in the feed, visually distinguishable from university events	Pass
TC008	Functional	Event filtering	Filter events list by category/date	Only matching events are returned	Pass
TC009	Functional	Register event interest	Tap 'Interested' on an event	Interest count increments; organiser can view attendee count/list	Pass
TC010	Functional	Cancel event interest	Cancel previously registered interest	Interest count decrements accordingly	Pass
TC011	Functional	Society activity post	Society representative posts an activity update	Update appears on the society's page and/or general feed	Pass
TC012	Functional	Society profile display	Open a society's profile page	Society information displays correctly and completely	Pass
TC013	Functional	Society sign-up	Student signs up to join a society	Sign-up is recorded; society rep can view interested/member list	Pass
TC014	Functional	Duplicate society sign-up	Student attempts to sign up for the same society twice	System prevents duplicate entry or shows already-joined status	Pass
TC015	Functional	Report lost item	Submit a lost item report with description and photo	Item appears in a searchable Lost & Found board	Pass
TC016	Functional	Search lost & found	Search the Lost & Found board by keyword	Matching reported items are returned	Pass
TC017	Functional	Resolve lost & found item	Mark a lost/found item as resolved	Item status updates and is	Pass

				removed from active search results	
TC018	Functional	Check room availability	View classroom availability for a given time slot	System shows correct available/occupied rooms	Pass
TC019	Functional	Book already-reserved room	Request a room that is already booked for that slot	Request is rejected or flagged as a conflict	Pass
TC020	Functional	Book available room	Request a room that is free for that slot	Booking is confirmed and reflected in the availability view	Pass
TC021	Functional	Cancel room booking	Cancel a confirmed booking	Room becomes available again for that slot	Pass
TC022	Functional	Request academic support	Submit a request for tutoring/study group/mentorship	Request is logged and routed to the relevant staff/coordinator	Pass
TC023	Functional	FAQ search - match	Search FAQs for a common query (e.g. 'exam timetable')	Relevant FAQ answer is returned	Pass
TC024	Functional	FAQ search - no match	Search FAQs for an unrelated/nonsense term	System shows a graceful 'no results' message	Pass
TC025	Functional	Authorised content edit	Authorised staff edits existing published content	Update is saved and reflected immediately, with timestamp/version	Pass
TC026	Functional	Unauthorised content edit	Non-authorised user attempts to edit published content	Edit action is blocked	Pass
TC027	Access Control	Student attempts to publish	Student account attempts to publish an announcement	Action is denied (student has view-only access)	Pass
TC028	Access Control	Academic staff publishes	Academic staff account publishes academic content	Action succeeds as expected for their role	Pass
TC029	Access Control	Cross-role restriction	Finance staff attempts an action outside their role (e.g. posting events)	Action is denied; role-based restriction enforced	Pass
TC030	Functional	View academic calendar	Open the academic calendar view	Correct exam periods, add/drop dates and milestones are displayed	Pass
TC031	Functional	Filter academic calendar	Filter calendar by semester/date range	Correct subset of dates is shown	Pass
TC032	Functional	Onboarding info access	New/first-year student opens onboarding section	Onboarding content is visible and easy to locate	Pass
TC033	Functional	Emergency alert delivery	Authorised staff sends an emergency/safety alert	Alert is delivered promptly and prioritised above normal content	Pass

TC034	Functional	Schedule change notice	Post a closure or schedule-change notice (e.g. weather closure)	Notice is clearly visible, flagged as urgent/time-sensitive	Pass
TC035	Functional	Submit feedback	Student submits feedback or a question	Submission is recorded and retrievable by staff	Pass
TC036	Functional	Volunteering visibility	Open the volunteering opportunities section	Opportunities are listed correctly with relevant details	Pass
TC037	Functional	Alumni engagement content	Post an alumni-related activity or opportunity	Content is visible to current students	Pass
TC038	Functional	Job/internship visibility	Open job/internship/placement listings	Listings display correctly with relevant details	Pass
TC039	Functional	Filter job listings	Filter job listings by type (job/internship/placement)	Only matching listings are returned	Pass
TC040	Functional	Report facility issue	Student reports a maintenance/facility issue	Report is logged with location and description	Pass
TC041	Functional	View facility issue queue	Maintenance staff opens reported issues list	Reported issue is visible in the staff queue	Pass
TC042	Functional	Staff directory search	Search the staff/departamental directory	Correct contact information is returned	Pass
TC043	Functional	Financial support info	Open scholarships/financial aid information	Correct, complete information is displayed	Pass
TC044	Functional	Sports & recreation info	Open sports/recreational facility information	Correct info displayed, including booking/access details	Pass
TC045	Functional	Dining information	Open canteen/dining menu and hours	Correct menu and operating hours are displayed	Fail
TC046	Functional	Printing services info	Open printing/copying/stationery service info	Correct information is displayed	Pass
TC047	Functional	List textbook for exchange	Student lists a second-hand textbook for exchange	Listing appears and is searchable by other students	Pass
TC048	Functional	Search textbook exchange	Search textbook exchange listings by title/course	Matching listings are returned	Pass
TC049	Functional	Guest lecture publicity	Publicise a guest lecture/industry talk	Item appears, visually distinct from regular campus events	Pass
TC050	Functional	Wellbeing support info	Open wellbeing/counselling support section	Correct support information is displayed	Pass
TC051	Functional	IT support info	Open IT/technical support information section	Correct support information is displayed	Pass

TC052	Functional	Library resources info	Open library resources/opening hours section	Correct information is displayed	Pass
TC053	Functional	Notifications	Notifications send to the relevant person	Relevant notifications are displayed correctly	Pass

12. Validation & Evidence

The prototype was validated by testing the main user journeys.

The objective was to demonstrate that the implemented features correspond to the selected business requirements.

13. Key Design Decisions

Decision 1 – Centralised Dashboard

Decision: Provide one main dashboard.

Reason: The problem statement identifies disconnected information channels as a key issue.

Decision 2 – Role-Based Access

Decision: Separate student and staff permissions.

Reason: Official university information should only be modified by authorized users.

Decision 3 – AI Assistant

Decision: Include an AI assistant.

Reason: Identifies conversational AI as a way to help students find information quickly.

Decision 4 – Modular Design

Decision: Separate major services into modules.

Reason: This makes the application easier to maintain and extend.

14. Development Process

The development process followed these stages:

Stage 1 – Problem Analysis

The team analyzed the problem statement and identified the main student information challenges.

Stage 2 – Requirement Identification

The team reviewed the business and non-functional requirements and selected the requirements that could realistically be implemented within the hackathon.

Stage 3 – Planning

Features were prioritized according to their Importance and implementation feasibility.

Stage 4 – UI/UX Design

The main screens and user journeys were designed.

Stage 5 – Development

The team implemented the core functionality.

Stage 6 – Integration

Individual features were integrated into the main application.

Stage 7 – Testing

The team tested the main features and corrected identified issues.

Stage 8 – Final Validation

The completed prototype was reviewed against the selected requirements.

15. GitHub & Team Collaboration

The project was developed collaboratively using GitHub.

The team used:

- Feature branches

- Commits
- Pull requests
- Code reviews
- Issue/task management

Each team member was responsible for specific areas while also supporting the integration and testing of other features.

16. Future Improvements

Future versions could include:

- Integration with university student records
 - Integration with official timetabling systems
 - Push notifications
 - Advanced AI capabilities
 - Real-time room availability
 - More advanced analytics for university administrators
 - Integration with university authentication systems
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17. Conclusion

One Uni powered by UCL was developed as a centralized platform to address the challenge of fragmented campus information.

The prototype brings multiple student services together in one platform and provides separate functionality for students and authorized staff.

The solution focuses on improving access to information, increasing visibility of campus opportunities and reducing the effort required to find common university information.
